Statistics for Data Science
Group [4] Project Report – Students' Academic Performance

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Abstract

This report investigates the factors influencing student academic performance, measured by Grade Point Average (GPA), through statistical and machine learning techniques, including Ordinary Least Squares (OLS) regression, propensity score matching, and Bayesian regression. Key findings reveal that absences significantly negatively impact GPA, while study time and parental support positively influence it. The analysis incorporates thorough validation using residual plots and performance metrics like R-squared and MSE. These findings highlight the critical role of attendance and parental support in enhancing academic performance, offering valuable insights for educators and policymakers.

1. Objectives and Hypothesis

1.1. Introduction - Problem Identification

What affects the academic performance of high-school students? Is it the number of hours spent studying? Is it correlated to their parents' academic background or support? Do extracurricular activities, music or sports correlate with academic success? Using the dataset here, we will try to answer some of these questions, via in-depth statistical analysis, as will be described in the following sections.

1.2. Data Set Description

This comprehensive dataset[1], sourced from Kaggle, provided information on the academic performance of 2,392 high school students. It included a wide range of data, covering demographics, study habits, parental involvement, extracurricular activities, and academic performance (GPA and grade class). A summary of variable definitions and value types is presented in **Table 1**. All variables, except for 'GPA', 'GradeClass', and 'ID', were defined as independent variables, whereas 'GPA' was defined as the dependent, target variable. 'GradeClass', a categorical variable derived from 'GPA' values (as shown in Table 1), was used as the target variable for specific statistical tests. 'ID' was utilized solely to identify duplicate entries.

1.3. Hypothesis

In order to investigate if there is a statistically significant relationship between our target variable, GPA, and the independent variables in the dataset, we formulated the following hypotheses:

1.3.1. General Hypothesis for Linear Regression

Null Hypothesis (H_0):

There is no linear relationship between the target variable, GPA, and any of the independent variables, meaning none of the predictors have a significant effect on GPA:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \cdots = \beta_i = 0$$

Alternate Hypothesis (H_1):

At least one regression coefficient is not equal to zero, indicating a significant correlation between at least one independent variable and GPA:

$$H_1: \exists\ \beta_0 \neq 0\ for\ at\ least\ one\ i$$

1.3.2. Hypothesis for Categorical Variables, One-way Analysis of Variance (ANOVA)

Investigating whether there are statistically significant differences in the mean GPA across different groups defined by the categorical variables.

Null Hypothesis (H_0):

All the means of all possible groups are equal:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$$

Alternate Hypothesis (H_1):

At least one group has a mean GPA different from the others:

$$H_1: \exists\ \mu_k \neq \mu_j\ for\ some\ j \neq k,$$ or simply $at\ least\ one\ \mu_k\ is\ not\ equal$

1.3.3. Hypothesis for Two Categorical Variables, Two-way ANOVA

Determining if the mean GPA is significantly different across the levels of two categorical variables and to see if there is an interaction effect between the two variables on GPA:

Null Hypothesis (H_0):

There is no interaction effect between factor A and B on GPA:

$$H_{0A}: \mu_{A1} = \mu_{A2} = \mu_{A3} = \cdots = \mu_{Ak}$$
$$H_{0B}: \mu_{B1} = \mu_{B2} = \mu_{B3} = \cdots = \mu_{Bl}$$
$$H_{0AB}: There\ is\ no\ interaction\ effect\ between\ factor\ A\ and\ factor\ B$$

Alternate Hypothesis (H_1):

There is an interaction effect between factor A and factor B

$$H_{1A}: At\ least\ one\ \mu_{Ak}\ is\ not\ equal\ (\text{Factor A})$$
$$H_{1B}: At\ least\ one\ \mu_{Bl}\ is\ not\ equal\ (\text{Factor B})$$
$$H_{1AB}: There\ is\ an\ interaction\ effect\ between\ factor\ A\ and\ factor\ B\ (Interaction)$$

#	Section	Column Name	Description	Dtype
1	**Student ID**	**StudentID**	Unique identifier (1001 to 3392)	int64
2	**Demographics**	**Age**	Student's Age: 15-18	int64
3		**Gender**	Student's Gender: 0-male, 1-female	int64
4		**Ethnicity**	Student's ethnicity: 0- Caucasian 1- African American 2- Asian 3- Other	int64
5	**Parental Education**	**ParentalEducation**	Parental education level: 0: None 1: High School 2: Some College 3: Bachelor's 4: Higher	int64
6	**Study Habits**	**StudyTimeWeekly**	Weekly study time, hours: 0-20	float64
7		**Absences**	Number of absences in a school year: 0 – 30	int64
8		**Tutoring**	Tutoring status: 0 – No, 1 – Yes	int64
9	**Parental Involvement**	**ParentalSupport**	The level of parental support: 0: None 1: Low 2: Moderate 3: High 4: Very High	int64
10	**Extracurricular Activities**	**Extracurricular**	Participation in extracurricular activities: 0 - No, 1 – Yes	int64
11		**Sports**	Participation in sports: 0 - No, 1 – Yes	int64
12		**Music**	Participation in music activities: 0 - No, 1 – Yes	int64
13		**Volunteering**	Participation in volunteering: 0 - No, 1 – Yes	int64
14	**Academic Performance**	**GPA**	Target Variable, Grade Point Average: 0.0 - 4.0	float64
15		**GradeClass**	Students' Classification based on GPA: 0: 'A' (GPA >= 3.5) 1: 'B' (3.0 <= GPA < 3.5) 2: 'C' (2.5 <= GPA < 3.0) 3: 'D' (2.0 <= GPA < 2.5) 4: 'F' (GPA < 2.0)	float64

Table 1.

Data set columns (variables).

1.4. Objectives

Our goal was to test the null hypothesis and determine if it could be rejected by identifying independent variables that significantly correlate with GPA. Statistical investigation included statistical regression models, causal inference, and Bayesian regression. Our rationale was that academic success is not random and is significantly influenced by specific factors. Hypothesizing that certain variables would show a strong correlation with academic performance, leading us to assume the null hypothesis will be rejected.

2. Methodology

Our statistical analysis was mostly inferencing analysis, primarily designed to understand the relationships between the variables, identifying significant predictors, testing the null hypothesis and estimating causal effects. However, prediction was also included in the analysis, aimed to help us to further understand the relationships between variables. Specifically, our analysis was comprised of the following steps:

a. **Data Preprocessing** – Defining variable types, handling missing values, duplicates, transforming variables if needed, removing outliers, and so on, ensuring all variables are ready for analysis.
b. **Exploratory Data Analysis** – Using descriptive statistics and basic correlation analysis. This step was designed to gather basic information on variable distribution, generating frequency histograms, pie

charts and box plots, scatter plots, and basic correlation patterns by plotting a correlation matrix heatmap. This would help us understand the basic data characteristics, distribution, and correlation patterns.

c. **ANOVA** – To identify significant differences in mean GPA across different levels of categorical variables. One-way and Two-way ANOVA were applied to examine the correlation of single or paired categorical variables with GPA. Continuous variables were categorized to perform ANOVA, allowing us to analyze significant differences across levels of these continuous variables

d. **Ordinary Least Squares (OLS) Regression** - OLS regression was used to quantify the relationships between the independent variables and GPA, after confirming OLS assumptions of linearity, independence, homoscedasticity, and normality. Further analyses included backward elimination, examining the coefficients, residual plots, comparisons of actual vs. fitted values for the target variable, and learning curves. Additionally, the data was analyzed for the power of test (probability of correctly rejecting the null hypothesis when it is false), and plotting feature importance.

e. **Chi-square, T-test and Logistic Regression using 'GradeClass' as the target variable** – As 'GradeClass' is a categorical classification of the target variable 'GPA', and was used for additional statistical analysis that are appropriate for categorical dependent variables.

f. **Causal Inference with Generalized Propensity Scores –** Identifying the causal effect of specific treatment variables, shown as variables of high feature importance, on the target variable, 'GPA'.

g. **Bayesian Regression –** Independent variables with high feature importance were chosen as predictors and 'GPA' as the target. The model was fitted using Bayesian regression. Plotting the residuals to assess model fit and the posterior predictive distribution to visualize the predicted values of the Bayesian model against the actual GPA distribution. This approach provides a probabilistic framework for understanding the uncertainty in the model parameters and predictions.

3. Results

3.1. Data Preparation

3.1.1. Data Quality

The Dataset was of excellent quality, with no duplicates detected in the 'ID' column, or null values in any of the variables. Since the value range of all variables is quite limited, a significant number of outliers were not expected. Indeed, applying IQR to detect and remove outliers resulted in zero values removed. All initial values were either integers or floats, easing the preprocessing of the dataset for analysis.

3.1.2. Variables Preparation

All variables were numerical, some representing true numerical data, such as 'Age', and some representing ordinal values, such as 'ParentalSupport'. The Ethnicity variable was originally assigned numeric values of zero to three, which was found to be more suitable for ordinal variables. In order to more correctly represent this variable, One-hot coding was used to generate variables assigned with the ethnicity name: ''Ethnicity_Caucasian', 'Ethnicity_African_American', 'Ethnicity_Asian', and 'Ethnicity_Other'. Possible multicollinearity resulting from this manipulation was evaluated later in the analysis.

3.2. Descriptive Statistics

3.2.1. Variable Distribution

First, variables were analyzed by visualizing frequency distribution. While most of our variables are of discrete values, the target variable, 'GPA', is a float and displayed normal distribution (**Appendix, Fig. S1**), with Skewness of 0.01 and Kurtosis of 2.1, indicating normal distribution. More elaborate investigation of normality is described in the following sections. An examination of the distribution of categorical variables, depicted in **Appendix, Fig. S2**, revealed several noteworthy observations. Gender distribution was fairly equal, with a slight majority for female students (51.1 female vs. 48.9% male). Parental education is mainly high school (30.4%) or some college (39.0%). In 'ParentalEducation', parents holding a Bachelor's or higher education comprised 15.3% and 5.0%, respectively, whereas most were high school graduates (30.4%) or had some college education (39.0%). Most students (69.9%) did not take tutoring lessons, and most did not participate in extracurricular activities (61.7%), particularly music (80.3% did not participate in any

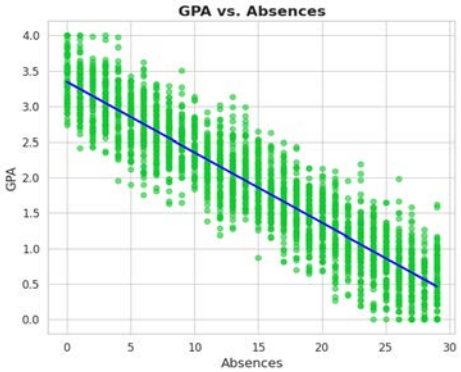


Fig. 1. GPA vs. Absences scatter plot.

music activities), or volunteering (84.6%). Parental support was mostly moderate to high (30.9% and 29.1%, respectively), or very high (10.6%), whereas the portion of parents providing low or no parental support is rather small (20.4% and 8.9%, respectively). To compute the pie charts, a new dataframe, where the numerical values were mapped with the categorical values was generated. Variables of binary or discrete nature were visualized using box-plots, as depicted in the code. Boc plots are particularly useful in visualizing large discriminations in variable distributions and outliers, none of which were observed. Continuous variables such as 'StudyTimeWeekly' and 'Absences' were plotted against the target variable, resulting in visible negative linear correlation between 'Absences' and 'GPA' (**Fig. 1**). Plotting the correlation matrix heatmap (**Fig. 2**) indicated a strong negative correlation between 'Absences' and 'GPA', with a Pearson correlation coefficient of r = -0.92. 'GPA' was also found to be weakly positively correlated with 'StudyTimeWeekly' (r = 0.18), 'ParentalSupport' (r = 0.19) and 'Tutoring' (r = 0.15).

3.2.2. Challenges in data preparation

There were no major challenges in preparing the dataset for analysis. All variables were assigned numeric values, which made the preprocessing and the exploratory data analysis pretty straightforward. As most of the variables display a binary or a small range of values, a careful examination was made later in the analysis to assure that there is no bias towards variables that are of a wider range of values.

3.3. ANOVA Analysis

Analysis of Variance (ANOVA) is a statistical method used to identify significant differences between the means of three or more groups. It is particularly useful for analyzing categorical variables, as it helps determine whether the differences between group means are statistically significant, indicating that the groups likely represent distinct populations. For categorical independent variables, groups are defined by the distinct levels or categories within the variable. Here, ANOVA was applied in three steps: (i) ANOVA For categorical independent variables against the target variable, 'GPA'; (ii) One-way ANOVA for the continuous independent variables 'StudyTimeWeekly' or 'Absences' after categorization, against the target variable, 'GPA; and (iii) Two-way ANOVA for 'StudyTimeWeekly' and 'Absences', and then for 'ParentalSupport' and 'Tutoring', against the target variable, 'GPA'. First, ANOVA was conducted for the

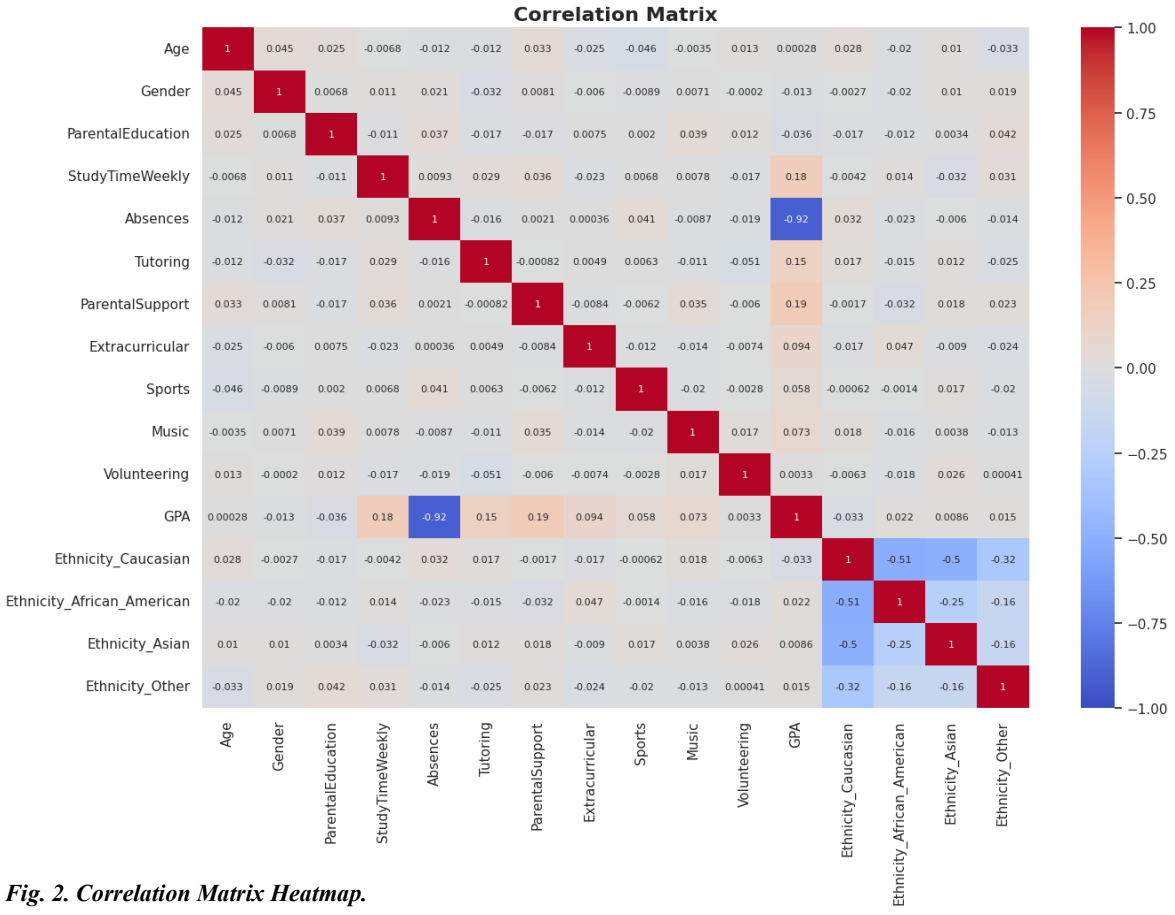


Fig. 2. Correlation Matrix Heatmap.

following categorical variables: 'Gender', 'ParentalEducation', 'Tutoring', 'ParentalSupport', 'Extracurricular', 'Sports', 'Music', 'Volunteering', 'Ethnicity_Caucasian', 'Ethnicity_African_American', 'Ethnicity_Asian', and 'Ethnicity_Other'. Independent variables for which ANOVA resulted in a p-value (PR (> F)) of above 0.05, indicating no significant difference in 'GPA' across different variable levels include the following: 'Gender', 'ParentalEducation', 'Volunteering', and all 'Ethnicity' variables (**Appendix, Table S1**). The variables 'Tutoring', 'ParentalSupport', 'Extracurricular', 'Sports', and 'Music' showed statistically significant differences in GPA means, as indicated by their p-values ≤ 0.05. This means these factors have a meaningful impact on GPA. Of note, the residual values were NaN, suggesting possible multicollinearity, which will be investigated further in the following section. Second, One-way ANOVA for the continuous variables 'StudyTimeWeekly' and 'Absences', after binning them into 3 levels: [0, 10, 20] for study time, and [0, 15, 30] for 'Absences', labeled 'Low', 'Medium' and 'High', respectively. Both ANOVA resulted in high significance, of p-value $\leq$ 1.50 × 10^-13 for 'StudyTimeWeekly', and p-value ≤ 0.001 for 'Absences', indicating a clear significant difference in GPA scores across variable levels. Thus, suggesting a strong correlation between absences and GPA, as well as between study time and GPA scores. Third, a Two-way ANOVA was executed to test the combined correlation of weekly study time and absences and GPA, and for tutoring and parental support. A p-value of 0.959 indicated no significant interaction between 'StudyTimeGroup' and 'AbsencesGroup'. Similarly, analyzing external means of support by Two-way ANOVA of parental support and tutoring did not result in significant interaction between variables in predicting GPA scores.

3.4. OLS Regression

3.4.1. Validation of OLS assumptions

OLS regression works to estimate the coefficients, denoted as beta in **Eq. 1**, of the independent variables in the model, representing the strength and direction of the relationship between the independent variable and the target, dependent variable, GPA. Coefficient (beta) estimation by linear regression is achieved in a way of minimizing the sum of squared differences between observed and predicted values, that is, OLS attempts to find the best-fitting line (or hyperplane in higher dimensions) through the data points by minimizing these residuals. As a starting point, all independent variables were evaluated for their linear correlation with the target variable.

Eq. 1. OLS linear regression equation

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} + \varepsilon_i$$

$$GPA = \beta_0 + \beta_1 \cdot Absences + \beta_2 \cdot ParentalSupport + \beta_3 \cdot StudyTimeWeekly + \beta_4 \cdot Tutoring + \beta_5 \cdot Extracurricular + \beta_6 \cdot Sports + \beta_7 \cdot Music + \cdots + \beta_n \cdot Factor\ n + \varepsilon$$

Where:

$\beta_1, \beta_2, \ldots, \beta_n - Coefficients\ for\ independent\ variables$

$\varepsilon - Error$

First, considering the independent variable types in the dataset:

- Numerical variables – 'Age', 'StudyTimeWeekly', and 'Absences'.
- Binary variables – 'Gender', 'Tutoring', 'Extracurricular', 'Sports', 'Music', 'Volunteering'.
- Ordinal – 'ParentalEducation', 'ParentalSupport'.
- Dummy variables – 'Ethnicity_Caucasian', 'Ethnicity_African_American', 'Ethnicity_Asian', 'Ethnicity_Other'.

And the target variable, 'GPA', a float. As is the case in multiple linear regression, perfect multicollinearity needed to be evaluated. Multicollinearity can result from dummy variables, scale differences resulting from including variables of different scales, and possible interactions between independent variables. Here, ordinal variables were remained unchanged to avoid multicollinearity and higher complexity resulting from generating dummy variables, as the numerical values clearly represent the different levels of the ordinal variables. Binary and numerical variables were also remained unchanged. Perfect multicollinearity in the ethnicity dummy variable was analyzed as part of validating OLS assumptions, as follows.

Validation that OLS assumptions are met:

a. Multicollinearity – The independent variables should not be perfectly linearly related. This is because high multicollinearity can mask and make it difficult to identify the individual effect of each independent variable. Multicollinearity was analyzed by calculating Variance Inflation Factors (VIF). All ethnicity

variable VIFs equaled Infinity, indicating perfect multicollinearity. Thus, the smallest ethnicity population was detected and removed from the dataframe, resulting in VIF values all well below ten.

b. Linearity - The correlation between independent variables and the dependent variable is linear, i.e., the effect of independent variables is linear and additive. To assess linearity, residual scatter plots were generated. For continuous variables such as 'StudyTimeWeekly' and 'Absences' random scattering was observed (**Fig. 3**). For binary and ordinal variables, the plots did not show distinct patterns or differences between groups. Thus, linearity is assumed to hold.

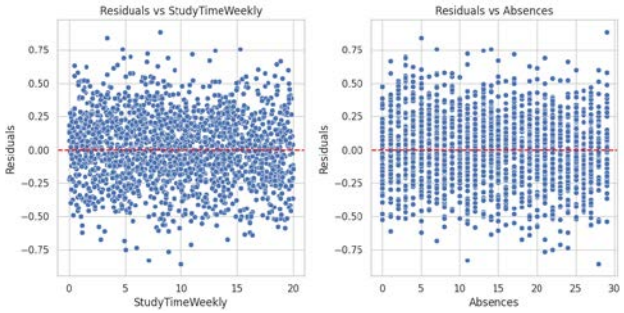


Fig. 3 Residual plots for Study Time Weekly and Absences variables.

c. Independence – Checking for autocorrelation in residuals using the Durbin-Watson statistics, which equaled 1.984, indicating no autocorrelation.

d. Homoscedasticity – Plotting residuals vs. fitted values, random scattering was observed, indicating Homoscedasticity, i.e., residuals variance is constant at every level of the independent variables.

e. Normality – By Q-Q plot, confirming normality of residual distribution. Residuals follow the straight line closely, thus normality assumption is satisfied (**Fig. 4**).

3.4.2. OLS goodness of fit

How well the model describes the observed data was evaluated by the following metrics: R-squared, Adjusted R-squared, and p-value (Prob (F-statistic)). R-squared and Adj. R-squared equaled 0.985, indicating excellent fit of the model, explaining about 98.5% of the variance in the dependent variable. The p-value, 0.00, suggests at least one independent variable is significantly related to the target variable, meaning the model is a good fit and has predictive power. This means the null hypothesis that all coefficients are equal to zero can be rejected with high confidence.

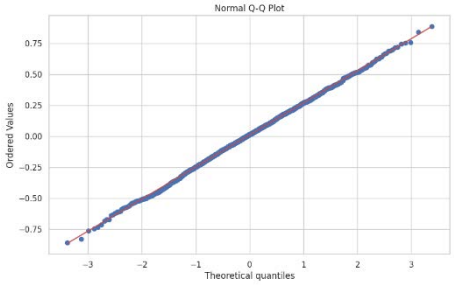


Fig. 4. Normality Q-Q Plot

3.4.3. OLS regression after scaling

Scaling the variables is important to ensure that each variable contributes equally to the model, regardless of their original scale. Using OLS for model prediction, the dataset was split into train (80%) and test groups (20%). Using the StandardScaler() from scikit-learn, the independent variables were transformed to have a mean of 0 and a standard deviation of 1, a process also known as z-score normalization. Standardization ensures that all features contribute equally to the model, regardless of their original scale., thus avoiding bias of variables with larger range. OLS regression without and after scaling resulted in R-squared = 0.954, Mean Squared Error (MSE) = 0.04 and p-value = 0.00, meaning that the OLS model is a good fit and has predictive power. Following standardization, 'Absences' was shown as the independent variable with the highest coefficient, with $\beta_{x3\ ('Absences')}$= -0.8442, meaning that an increase of one standard deviation in absences is associated with a decrease of 0.8442 standard deviations in GPA, all else being equal. Without scaling, 'Tutoring' $\beta_{Tutoring}$ = 0.2583, meaning given the low p-value of 0.00 for this variable, that students who receive tutoring are expected to have, on average, a GPA that is 0.2583 points higher than students who do not receive tutoring, assuming all other variables in the model are held constant. Grouping the variables 'Absences', 'StudyTimeWeekly', and 'Age' into three categories (ordinally, by binning into 0, 1, or 2), and re-running the OLS model, resulted in R-squared=0.802, MSE = 0.17, and p-value=0.00, indicating good fit of the model. Still, absences were shown to be an important factor in explaining GPA, with $\beta_{Absences_Level} = -0.7675$. Selecting only independent variables with $\beta \geq |0.05|$, namely, 'StudyTimeWeekly', 'Absences', 'Tutoring', 'ParentalSupport', 'Extracurricular', 'Sports', 'Music' resulted in the same R-squared, MSE and p-value values as for including all independent variables, suggesting they are redundant and non-significant for predicting GPA, which is certainly the case for example for all ethnicity variables with high p-value. Thus, these additional variables do not provide additional predictive power or explanatory information.

3.4.4. Variable Selection Using Backwards Elimination

Another method to identify significant variables for the model is backward elimination. Starting with the full model, the summary output was analyzed and identified 'Age', 'Gender', 'Volunteering', 'Parental Education', and all 'Ethnicity' variables as insignificant variables with p-values above 0.05 (**Fig. 5**). After removing these variables, the remaining variables in the model showed strong significance, with no improvement in R-squared or the probability associated with the F-statistic observed. However, there was an increase in the F-statistic and a slight improvement in the Akaike Information Criterion (AIC), where in the full model AIC equaled -772.3, whereas in the reduced model AIC = -781.4. Applying backward elimination procedure to the scaled model, yielded similar results. The significant variables remained consistent, reinforcing the robustness of the findings.

3.4.5. Power of the test

Estimating the power of the statistical test, defined as 1- β, is the probability of correctly rejecting the null hypothesis when it is false. Here, the estimated power of the test equaled 1.0 and calculated p-values for 1000 simulations were mostly well below 0.05, indicating the model is very good at detecting effects and there's a very low probability of a Type II error (**Fig. 6**). Majority of p-values were clustered near zero, indicating that in most simulations, the coefficients were found to be statistically significant. The robustness of the model was indicated by the narrow spread of p-values, mostly clustered towards zero. This consistency across simulations suggests that the model's findings are not sensitive to variations in the data, implying that the relationships between the independent variables and the dependent variable are stable.

```
### Without Scaling ###
Removing variable Ethnicity_African_American with p-value 0.8543702843421989
Removing variable Ethnicity_Asian with p-value 0.9352347322351193
Removing variable Volunteering with p-value 0.6700012563695046
Removing variable Ethnicity_Caucasian with p-value 0.24220716502117942
Removing variable Gender with p-value 0.24256738184752644
Removing variable Age with p-value 0.16325362720837008
                            OLS Regression Results
==============================================================================
Dep. Variable:                    GPA   R-squared:                       0.954
Model:                            OLS   Adj. R-squared:                  0.954
Method:                 Least Squares   F-statistic:                     5649.
Date:                Mon, 05 Aug 2024   Prob (F-statistic):               0.00
Time:                        13:16:38   Log-Likelihood:                 398.70
No. Observations:                1913   AIC:                            -781.4
Df Residuals:                    1905   BIC:                            -737.0
Df Model:                           7
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          2.5193      0.016    161.981      0.000       2.489       2.550
StudyTimeWeekly  0.0291      0.001     36.775      0.000       0.028       0.031
Absences      -0.0995      0.001   -187.392      0.000      -0.101      -0.098
Tutoring       0.2580      0.010     26.293      0.000       0.239       0.277
Extracurricular  0.1903      0.009     20.502      0.000       0.172       0.208
Sports         0.1856      0.010     18.965      0.000       0.166       0.205
Music          0.1521      0.011     13.455      0.000       0.130       0.174
ParentalSupport  0.1476      0.004     36.624      0.000       0.140       0.156
==============================================================================
Omnibus:                        5.236   Durbin-Watson:                   2.004
Prob(Omnibus):                  0.073   Jarque-Bera (JB):                4.971
Skew:                          -0.088   Prob(JB):                       0.0833
Kurtosis:                       2.824   Cond. No.                         70.1
==============================================================================
```

Fig. 5. Variable Selection Using Backwards Elimination via OLS.

3.4.6. Distribution of fitted vs. actual values

Plotting the actual values vs. the fitted values resulted in significant overlap (**Fig. 7**). Both histogram bars and KDE lines for actual and predicted values largely overlap, indicating that the model's predictions closely match the actual data distribution. This suggests that the model has good predictive accuracy. The means of the actual and predicted GPA values are overlapping, further confirming that the model predictions are well-centered around the actual values. Visually, the plot suggests that the model does not have significant bias, as the actual and predicted distributions align closely. If there were substantial deviations between the two distributions, it could indicate areas of overfitting or underfitting.

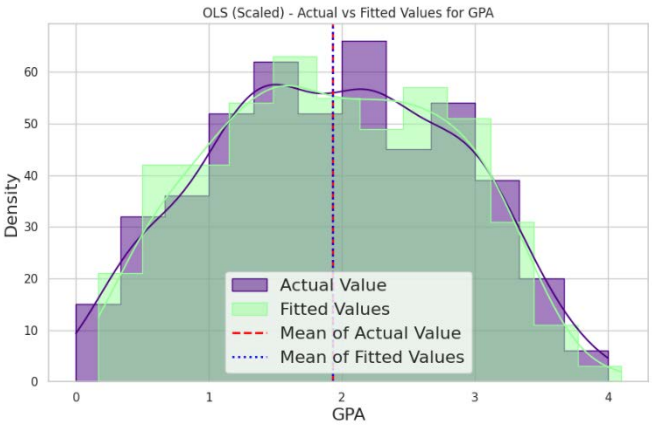


Fig. 7. OLS GPA Fitted vs. actual values distribution.

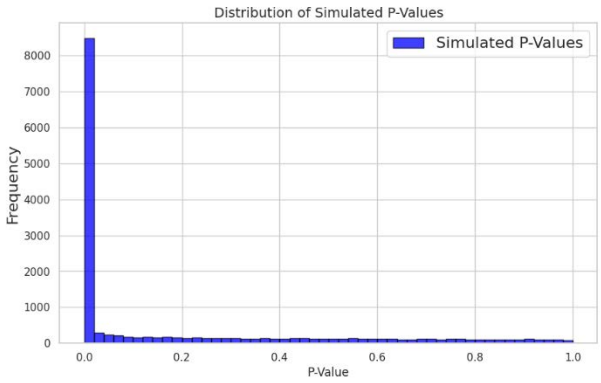


Fig. 6. Power of test.

3.4.7. OLS Learning curve

Plotting the learning curve shows the performance of the OLS model (scaled) on both the training and cross-validation datasets as the number of training examples increases (**Fig. 8**). The training score represents how well the model fits the training data. Here, the training score starts lower and increases as the number of training examples increases. With more data, the model can better generalize and fit the training set. It reached a plateau at around 600 observations. The cross-validation score represents the model's performance on unseen data, providing an indication of how well the model generalizes new data. Initially, the values are slightly higher than the training score, possibly due to the variability in smaller data samples. As the number of samples increases, it is stabilized and reaches a plateau at roughly 800 samples. The convergence of the training and cross-validation scores as the number of training examples increases suggests that the model is not overfitting the data. The scores plotted on the y-axis represent the negative mean squared error (MSE). The shaded areas around the lines represent the standard deviation of the scores. For the training score, the variance decreases as the number of training samples increases, whereas cross-validation score variance is relatively stable, suggesting consistent performance across the different cross-validation folds. Overall, the convergence of training and cross-validation scores indicates that the model is well-fitted, not overfitting the training data.

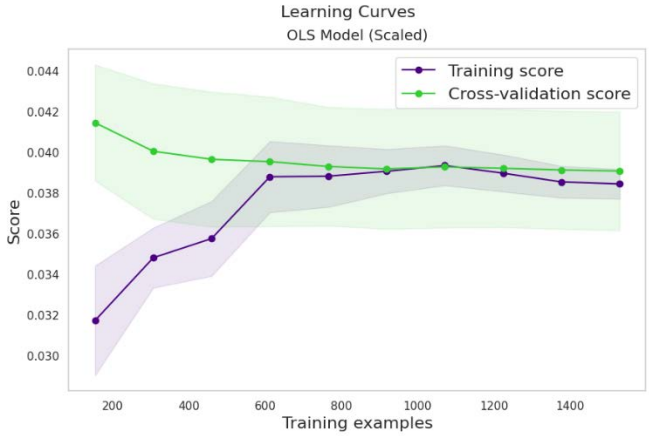


Fig. 8. OLS Learning curve.

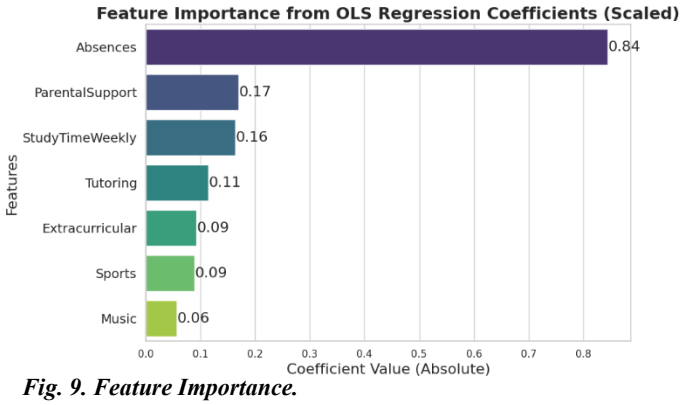


Fig. 9. Feature Importance.

3.4.8. Feature Importance

Plotting feature importance, it was apparent absences is the most important feature in explaining GPA (**Fig. 9**). The coefficients were represented in absolute terms, indicating the strength of the relationship between each independent variable and the dependent variable, 'GPA', after standardization. Parental support, tutoring and weekly study time are also important in explaining GPA. A similar trend was observed when plotting feature importance using the grouped absences, study time and age variables, grouped by levels, thus indicating variable values do not affect feature importance. Interestingly, demographic factors such as ethnicity, gender, and age show minimal to no impact on GPA in this analysis. Similarly, volunteering and parental education appear to have negligible effects.

3.4.9. Plotting Predicted vs. Actual GPA Values and Residuals Distribution

Plotting actual vs. predicted GPA values, based on the OLS scaled analysis, where each point on the horizontal axis represents a data point indexed by its position in the training data, revealed considerable overlap between predicted and actual values (**Fig. 10**), indicating an excellent fit of the model with original values. Plotting the residuals, representing the difference between predicted and actual GPA values, revealed random distribution around zero, indicating no systematic patterns in model errors.

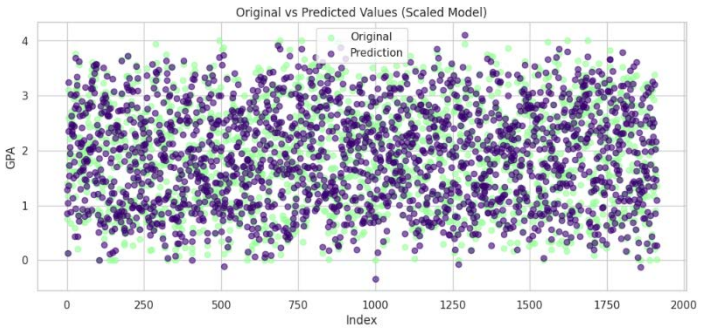


Fig. 10. Predicted vs. Actual GPA Values.

The residual density plot exhibits a normal distribution centered around zero, indicating that the model's errors are normally distributed (**Fig. 11**).

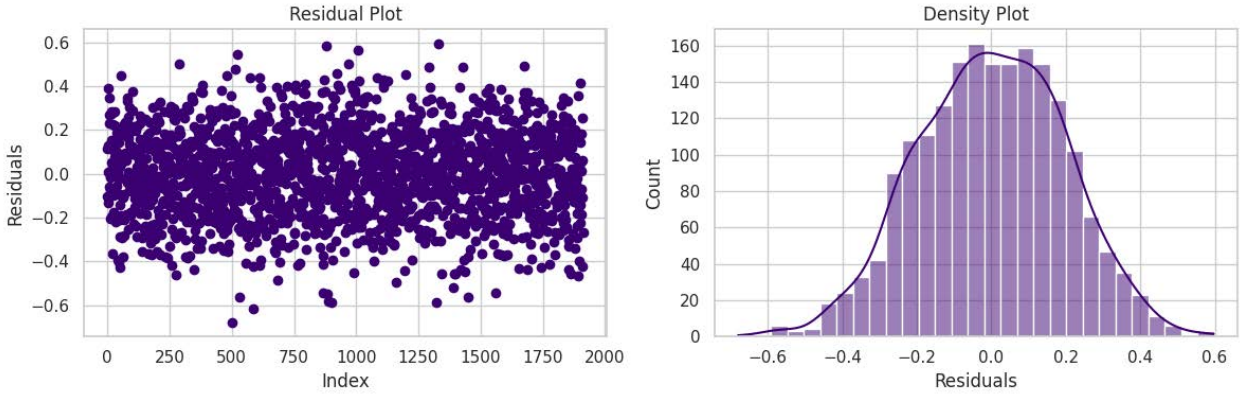


Fig. 11. Residuals scatter and residual density plots.

3.5. 'GradeClass' as categorical 'GPA'

3.5.1. 'GradeClass' as a categorical target variable

As 'GradeClass' is the categorical expression of 'GPA' it was used for Chi-square and t-tests with GradeClass as the target variable. Of note, 'GradeClass' is inversely correlated with 'GPA', where low 'GradeClass' are the categorical representation of high GPA and vice versa. Plotting Grade class for weekly study time and parental support clearly showed a correlation between higher study time and parental support and lower grade class, i.e., higher GPA. The opposite trend was observed in plotting absences, where higher absences were clearly concentrated in the grade classes representing lower GPAs. Grade class did not vary by gender, sports, or ethnicity group. Grade class zero, representing GPA≥ 3.5, was proportionally larger for students with tutoring, extracurricular activities, or high parental support (**Appendix, Fig. S3**).

3.5.2. Chi-square test for independence between categorical variables and GradeClass

This test checks whether there is a significant association between the categorical variable, 'GradeClass', and the independent categorical features. Looking at the p-values for each test, tutoring, extracurricular, parental support are significantly associated with grade class, meaning the distribution of grade class levels varies significantly across the categories of the independent feature, with $p - value \leq 0.05$.

3.5.3. Student's t-test for GradeClass and numerical variables

For numerical variables, the t-test was employed, comparing the means of numerical variables, 'StudyTimeWeekly' and 'Absences', across different 'GradeClass' groups. The ttest_ind() function used here performs an independent two-sample t-test, assuming the two groups are independent and that the populations from which the samples are drawn are normally distributed with equal variances. For weekly study time, there are notable differences in study time between most 'GradeClass' groups, particularly between the highest (0) and lower groups (2, 3, 4). This suggests that study time may be a key differentiator in student performance. The pattern is even more pronounced for absences, with clear and significant differences across most groups, indicating absenteeism as a strong differentiator of student performance, with higher-performing students attending school more regularly.

3.5.4. Logistic Regression with 'GradeClass' as an ordinal variable

Logistic regression was used to predict the likelihood of a student belonging to a particular grade class based on several independent variables. The independent variables were standardized using StandardScaler() and the multinomial logistic regression model was trained using the lbfgs solver, which is suitable for multiclass problems. The max_iter=1000 ensures the model has enough iterations to converge. The overall accuracy was 0.68, indicating the model correctly predicted the grade class for roughly 68.5% of the students in the test set. Highest precision was found in predicting grade class 4.0, with precision of 0.84, indicating that 84% of the instances predicted as class 4 are correct. Recall measures how many of the actual instances in each class were correctly identified. The highest recall was for grade class 4.0, with recall = 0.96, indicating that the model correctly identifies 96% of the actual instances of class 4. F1-Score is the harmonic mean of precision and recall. The higher the F1-score, the better the model's performance. Here, F1-score was highest for grade class 4.0, with F1-score = 0.90, and particularly low for grade class 0.0, with F1-

score=0.00, indicating poor model performance for this class. This indicates that the logistic regression model can benefit from further tuning, more data, or feature engineering, to improve performance for some of the grade class groups. Class imbalance can also be addressed by techniques such as adjusting class weights, but as the scope of this report is limited, this path was not pursued further.

3.6. Causal Inference with generalized propensity scores

To analyze the causal effect of the most influential independent variable in explaining 'GPA', 'Absences', causal inference was applied. Unlike correlation, which measures the association between variables, causal inference aims to establish whether changes in one variable, defined as a cause, directly lead to changes in the target variable, defined as the effect. The propensity score is the probability of receiving a particular treatment given a set of observed covariates, by summarizing these covariates into a single score, which can then be used to match or weight cases to reduce bias. The covariates used here to calculate the propensity score are those that are thought to possibly affect both treatment and outcome, i.e., potential confounders, who are of low p-value and high feature importance. At a low absence level, the estimated treatment effect is 1.496, meaning that on average, students with low absences have a GPA that is approximately 1.496 points higher than students with medium or high absences, after accounting for the propensity scores. At high absence level, estimated treatment effect was -1.448, meaning that students with high absences have a GPA that is approximately 1.448 points lower than students with low or medium absences. These trends generally remain unchanged when taking all independent variables into account.

3.7. Bayesian Regression

Finally, Bayesian regression analysis was applied, with 'Absences', 'ParentalSupport', 'StudyTimeWeekly', 'Tutoring' as independent variables. After standardization the model was fitted by Generalized Linear Model (GLM) with a Gaussian distribution, basically meaning linear regression where the target variable is assumed to follow normal distribution. 'Absences' coefficient $\beta_{Absences} = -0.841$, suggesting higher absences associated with lower GPA. Positive coefficients for 'ParentalSupport', 'StudyTimeWeekly' and 'Tutoring', of $\beta_{ParentalSupport} = 0.17$, $\beta_{StudyTimeWeekly} = 0.162$, and $\beta_{Tutoring} = 0.115$, respectively, means positive correlation between these and higher GPA. Residuals were randomly distributed around zero, indicating no obvious pattern and good fitting of the model. The posterior predictive distribution plot, representing the distribution of the predicted values (posterior predictions) from the Bayesian model showed a fairly normal spread, that overlaps the actual values very well (**Fig. 12**).

4. Conclusions

4.1. Hypothesis evaluation

Consistent results gathered from various analyses including OLS regression, logistic regression for 'GradeClass', propensity score analysis, and Bayesian regression all support the rejection of the null hypothesis, concluding that at least one of the regressors had a significant linear relationship with GPA. To ensure the validity of our findings, we performed the following steps:

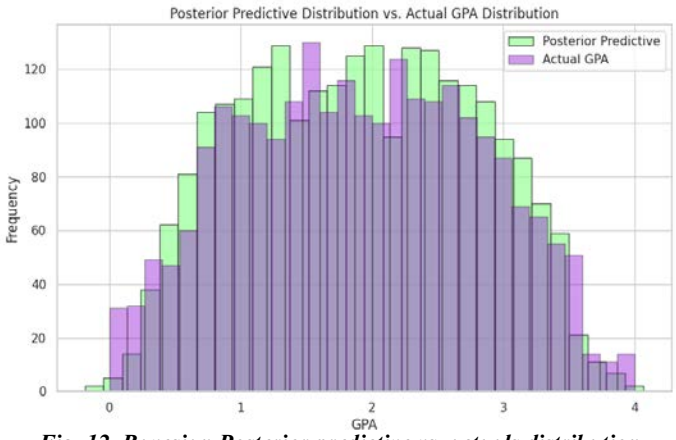


Fig. 12. Bayesian Posterior predictive vs. actuals distribution.

- **Scaling Regressors**

To eliminate the possibility of regressors with large scales having a disproportionate impact on the model, all the regressors were scaled to have a mean of 0 and a standard deviation of 1 using the StandardScaler() function. Re-running the OLS model with the scaled data yielded the same conclusion: at least one independent variable had a significant linear relationship with GPA.

- **Backward Elimination**

Employing a backward elimination approach, removing variables with the highest p-values step-by-step. This process was repeated with both unscaled and scaled data. Each model iteration retained a p-value of 0.00, reinforcing the decision to reject the null hypothesis. The final significant variables included 'StudyTimeWeekly', 'Absences', 'Tutoring', 'Extracurricular', 'Sports', 'Music', and 'ParentalSupport'.

- **Categorical Grouping of Continuous Variables**

Categorizing continuous variables ('Absences', 'StudyTimeWeekly', and 'Age') into categorical levels and retesting the hypothesis resulted in a p-value of 0.00, providing strong evidence to reject the null hypothesis.

- **ANOVA for Categorical Variables**

Testing if the mean GPA was equal across all groups for several categorical variables resulted in significant differences in GPA means for 'Tutoring', 'ParentalSupport', 'Extracurricular', 'Sports', and 'Music' variables. Grouping and testing continuous variables such as 'StudyTimeWeekly' and 'Absences' also resulted in rejecting the null hypothesis, indicating significant differences in GPA across groups.

Therefore, we confidently refute the null hypothesis, concluding that there is a significant relationship between the independent variables and the target variable, 'GPA'. Results clearly demonstrate that variables such as 'Absences', 'ParentalSupport', 'StudyTimeWeekly' have a statistically significant impact on GPA.

4.2. Factors correlated with academic success

OLS regression resulted in significant coefficients for 'Absences', 'ParentalSupport' and 'StudyTimeWeekly', with p-value ≤ 0.05. Specifically, the extremely low p-value for 'Absences' suggest a strong negative correlation to 'GPA'. Furthermore, causal inference analysis using propensity scores indicated a significant treatment effect of absences on GPA. The observed differences in GPA across different levels of 'Absences' (Low, Medium, High) were significant, with p-values indicating these differences were unlikely to be due to chance. The Bayesian model also indicated that the predictors significantly affect GPA, with the estimates being precise and reliable, with pseudo R-squared of 1.00, and p-values of 0.000 for all predictors. Logistic regression showed significant relationships between predictors and 'GradeClass' categories, particularly 'StudyTimeWeekly' and 'ParentalSupport'. The model's accuracy and classification metrics (precision, recall, F1-score) further indicate that predictors are not independent of the outcome. Thus, variables such as 'Absences', 'ParentalSupport', and 'StudyTimeWeekly' have a statistically significant impact on 'GPA'.

4.3. Limitations and challenges

Data accuracy, such as self-reported measures or inconsistencies in data recording, can impact the analysis. Undetected multicollinearity can also affect the analysis. If not all sources of multicollinearity are detected, unreliable coefficient estimates can result. Furthermore, there may be other variables not included in the analysis that could influence both the predictors and the outcome, potentially confounding the results. While the analysis, including the use of propensity scores, attempts to control for confounding, it does not establish causality definitively. The results indicate associations but do not unequivocally prove that changes in the predictors cause changes in GPA. Finally, findings are specific to the dataset used, and if not representative of a larger population, the results may not generalize well beyond the specific sample analyzed.

4.4. Ideas for future analysis

Exploring non-linear relationships, interaction effects between variables, and including additional predictors, such as location or socioeconomic data, could enhance the analysis. Applying advanced machine learning techniques might capture more complex patterns and improve prediction accuracy. A more detailed analysis at the subject level could reveal if significant variables consistently impact subjects like math, language, or science. Once these relationships are established, the findings can be leveraged to promote academic success. Investing in programs that support extracurricular activities or tutoring can foster student growth and contribute to economic advancement.

References

[1] R. El Kharoua, 2024, https://www.kaggle.com/datasets/rabieelkharoua/students-performance-dataset/data.